Step 4: Weather to power (ticket PPA-4)

> **Time needed:** about 5 hours. Concepts take 90 minutes, code tour and findings 60 minutes, hands-on, notebook and exercises 2 hours, to-dos 30 minutes. **You will be able to:** explain where long-term weather data comes from, turn irradiance and wind into the hourly output of a real solar and wind park, compute P50 and P90 volumes, and check a model when no official number exists to tie it out to.

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0. The scenario

Thursday. The buyer's bank joins the call:

> *"Your pay-as-produced price looks fine, but we lend against energy, not prices. What does this park produce in a normal year, what in a bad one, and how sure are you? We want P50 and P90, from your own site, not from a national average. Ticket PPA-4."*

Until now the parks were the national generation profile, scaled (Step 3). That was good enough to compare contract structures, but it hides two things a lender and a pricing analyst care about: the **site's own weather** (a single park is not the whole fleet), and **how much a year can differ from another** (volume risk). Both need a long weather record at the exact location and a model that turns weather into power.

Ticket PPA-4: acceptance criteria (all met today)

- ☑ ERA5 weather for two real sites, 1995 to date, cached with provenance (grid point, licence, completeness).
- ☑ The timestamp convention handled and verified against the sun's position (test and a check on every build).
- ☑ A pvlib solar park (50 MWp) and a windpowerlib wind park (30 MW) from configuration.
- ☑ Annual energy per weather year, P50 and P90 (one year and ten years), trend.
- ☑ Checks against national generation: hourly correlation, monthly shares, capture rate.
- ☑ Report, learning page, notebook section, exercises.

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1. Concepts you need today

1.1 Volume risk, the third risk

Step 2 and Step 3 were about **price** and **shape**. The third risk is **volume**: how much energy the park produces. A pay-as-produced PPA moves price and shape risk to the buyer, but the producer still sells *whatever the weather delivers*: a dull year means fewer MWh at the same fixed price. Lenders size debt on a volume they are confident about (P90 or P99), and a PPA offer with a delivery guarantee must be priced on it.

1.2 Where long-term weather data comes from: reanalysis

A park rarely has decades of on-site measurements. The industry standard is a **reanalysis**: a weather model run backwards in time that continuously assimilates the observations available (stations, satellites, radiosondes, aircraft). The result is a physically consistent, gap-free record everywhere on Earth.

- **ERA5** (ECMWF, for the EU's Copernicus Climate Change Service) is the reference reanalysis: hourly, from 1940, on a grid of about 0.25° (here about 28 × 16 km).
- We use it through the **Open-Meteo archive API**: free, no key, CC BY 4.0. It snaps the request to the nearest grid point and tells you which one it used.
- A grid value is an **average over the cell**. Local effects (a hill, a forest edge, the coast) are smoothed out. Section 4.2 shows how much that matters.

> **Real-world note:** yield consultants combine a short on-site measurement campaign with 20-30 years of reanalysis: "measure, correlate, predict" (MCP). Encavis calls its long-term reference "standard weather" based on 30 years. That is exactly the role our 31 ERA5 years play.

1.3 Solar: from irradiance to AC power

Irradiance arrives as three components (all in W/m²):

Component	Meaning
GHI, global horizontal	everything falling on a flat surface
DNI, direct normal	the beam from the sun's disc, on a surface facing the sun
DHI, diffuse horizontal	sky light scattered by clouds and air

The chain in `weather/pv.py` (pvlib does each step):

1. **Sun position** for each hour, taken at **mid-interval** (an hourly mean of 10:00-11:00 belongs to the sun at 10:30).
2. **Plane-of-array (POA) irradiance:** the panels are tilted 25° towards the south, so GHI/DNI/DHI are transposed onto that plane (Hay-Davies model).
3. **Cell temperature** (Faiman model): hot cells produce less, about **–0.35% per °C**.
4. **DC power** (PVWatts): proportional to POA irradiance, corrected for temperature.
5. **System losses** of 10%: soiling, mismatch, wiring, availability.

6. **Inverter** (96% efficiency) with a **DC/AC ratio of 1.2**: 50 MWp of panels on 41.7 MW of inverters. On the sunniest hours the inverter **clips** the output.

The result per year divided by the DC capacity is the **specific yield** (kWh per kWp). German ground-mounted parks reach roughly 950-1,150 kWh/kWp; the east (Lusatia) is among the sunnier regions.

1.4 Wind: from wind speed to MW

- **Wind shear:** wind gets stronger with height. ERA5 gives the speed at 10 m and at 100 m, so we can measure the shear each hour with the power law `alpha = ln(v100 / v10) / ln(100 / 10)` and extrapolate to the hub: `v_hub = v100 · (hub height / 100)^alpha`. Alpha is clipped to [0, 0.6].
- **Power curve:** a turbine produces nothing below the **cut-in** speed (about 3 m/s), rises roughly with the **cube** of wind speed, reaches **rated power** (3.45 MW at 12 m/s for the Vestas V126) and shuts down above **cut-out** (22.5 m/s in the library curve).
- **The cube law means the mean wind speed is not enough.** Two sites with the same mean but a different distribution produce different energy. That is why we model every hour.
- **Losses** of 12%: wake effects of turbines on each other (~7%), availability (~3%) and electrical losses (~2%).
- Energy per year divided by capacity gives the **full-load hours**: how many hours at full power would produce the same energy. The German onshore fleet averages below 2,000; modern large-rotor turbines at good sites in the north reach 3,000 and more.

1.5 P50 and P90

From the annual energies of N weather years (mean μ, standard deviation σ):

```
P50          = the median year: exceeded in half of all years
P90 (1 year) = μ − 1.2816 σ        exceeded in 90% of years (normal approximation)
P90 (10 yr)  = μ − 1.2816 σ / √10   for the average over ten years
```

- 1.2816 is the 90% quantile of the standard normal distribution.
- The **10-year** P90 is much closer to the mean: good and bad years average out. A 10-year PPA or loan cares about the 10-year number.
- It assumes **independent years**. If weather has trends (solar "brightening") or persistent periods, the true uncertainty is larger (exercise 4).
- Lenders often size debt on **P90 or P99**; the budget uses **P50**.

1.6 Checking a model with no official number

In Step 1 and Step 2 we reconciled against published figures. Nobody publishes the output of an illustrative park. Instead we check that the model **behaves like reality**:

1. **Benchmarks:** specific yield and full-load hours in the right range for the region and technology.
2. **Pattern:** hourly correlation with the national fleet of the same technology (Step 1 data). High for solar (everyone sees the same sun), lower for one wind site.
3. **Seasonality:** the share of annual energy per month should look like the fleet's.

4. **Economics:** the site's capture rate at real prices (Step 2) should be close to the national one, with explainable differences.
5. **Timestamps:** irradiance must line up with the sun (section 4.1).

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2. What was built: architecture

```
Open-Meteo archive API (ERA5)                          config/project.toml
  GHI, DNI, DHI, T2m, wind 10/100 m, hourly            [pricing.parks.*] site + model
            │                                           [weather] first year, pause, licence
            ▼                                                       │
data/open_meteo.py   OpenMeteoClient.hourly() — parse_hourly_payload() (pure)
            │                                                       │
data/weather_pipeline.py  fetch_weather(): one raw file per site and year (+ manifest)
            │             build_generation(): to_interval_beginning() — sun check
            ▼
weather/pv.py (pvlib)          weather/wind.py (windpowerlib)
            └──────────────┬──────────────┘
                           ▼
data/processed/weather_<park>.parquet · generation_<park>.parquet (mw = MWh per hour)
                           │
weather/yields.py  annual_energy, p_values      analysis/generation.py  trend, checks
                           ▼
analysis/step04_report.py → reports/step04_generation.md + 2 charts   (`ppa-lab weather`)
```

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3. Code tour

3.1 `data/http.py`: one polite client for every source

The retry logic from Step 1 (timeouts, exponential back-off, obeying `Retry-After`) lived inside the Energy-Charts client. A second source would have copied it, so it moved into `JsonApiClient`; `EnergyChartsClient` and `OpenMeteoClient` are now ten-line subclasses with their own base URL and error type. Step 1's client tests passed unchanged: that is what makes a refactor safe.

3.2 `data/open_meteo.py`: parse, convert, verify

- `parse_hourly_payload` is pure: it checks units (forgetting `wind_speed_unit=ms` would give km/h, and the parser refuses it), keeps the API's own time labels (the **raw** layer stays what the source said) and returns the ERA5 grid point used.
- `to_interval_beginning` converts to the project convention: radiation labelled at the **end** of its hour moves back one hour; instantaneous wind and temperature become the mean of the two instants around the hour. It refuses an index with gaps.
- `radiation_timing_offset` compares irradiance with the clear-sky sun at −30, +30 and +90 minutes after the row label and returns the best fit. For correct data it is +30.

3.3 `data/weather_pipeline.py`: cache with provenance

One raw Parquet file per **site and year** with a manifest: request parameters, the ERA5 grid point and its elevation, the licence, and `complete`. ERA5 is published with a delay of about five days, so the current year is marked incomplete and re-downloaded on the next run; finished years come from the cache. Changing a park's coordinates triggers a re-download for that park only (section 4.2 used exactly that).

`build_generation` concatenates the years, converts the timestamps, **runs the sun check for the solar site and stops the build if it fails**, runs the site model and writes `weather_<park>.parquet` and `generation_<park>.parquet`.

3.4 `weather/pv.py`, `weather/wind.py`, `weather/yields.py`

Each model is one function from a weather frame to an MW series, with all parameters from the configuration (`tilt_deg`, `dc_ac_ratio`, `turbine_type`, `hub_height_m`, `losses_pct`, ...). The wind power curve comes from windpowerlib's turbine library, which ships with the package (no download in CI). `yields.py` turns hourly MW into annual energy per **complete** local year and computes P50/P90.

3.5 Tests: what each one protects

Test file	Protects against
`test_open_meteo.py` (7)	wrong units; radiation not shifted (caught by two tests, one on real data); gaps; wrong request parameters
`test_weather_models.py` (9)	sun output at night; exceeding the inverter limit; wrong shear maths; power curve beyond cut-out; no temperature loss
`test_yields.py` (2)	partial years counted as weather years; wrong P90 formulas
`test_weather_pipeline.py` (4)	re-downloading cached years; never refreshing the current year; a shifted dataset slipping through
`test_generation_analysis.py` (3)	plain instead of weighted capture price; UTC months; trend per year instead of per decade
`test_step04_report.py` (2), `test_cli.py` (+1)	the report and `ppa-lab weather` end to end, including too-short ranges

The key tests were checked by planting the bug they protect against and watching them fail.

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4. Four real findings

4.1 The weather data is labelled at the end of the hour

- **Symptom:** none yet. This was found *before* it could cause one, by reading the source's documentation and then testing the claim with data.
- **Evidence:** compare hourly irradiance with the clear-sky sun computed for different times. A full year (2025) of converted data:

Sun position taken at	Correlation with irradiance
30 min before the row label	0.872
30 min after (mid-interval)	**0.919**
90 min after	0.890

- **Why it matters:** without the shift, every solar MWh sits one hour late. Solar output would meet the evening price ramp instead of the midday dip, and the capture price would come out too high, exactly the error that costs money in a PPA.
- **Guard rails:** a unit test on a real two-day ERA5 sample, and a check inside every build that stops `ppa-lab weather` if the data ever arrives shifted (`test_build_generation_refuses_weather_with_shifted_timestamps`).

4.2 The first wind park was in the sea

- **Symptom:** the first run gave the wind park **3,961 full-load hours**, an offshore value.
- **Investigation:** the report's site table showed the ERA5 grid point for 54.50 N, 9.00 E at **0 m elevation**: the cell next to Husum is partly Wadden Sea. Five candidate cells, one year of wind at 100 m each:

ERA5 grid point	Elevation	Mean wind at 100 m (2024)
54.50 N, 9.00 E (first choice)	0 m	7.52 m/s
54.50 N, 9.25 E	**30 m**	**7.11 m/s**
54.50 N, 9.50 E	24 m	6.88 m/s
54.75 N, 9.25 E	22 m	7.01 m/s
54.25 N, 9.25 E	29 m	6.98 m/s

- **Fix:** the park moved one cell (about 16 km) inland, to a land cell. Result: **3,668 full-load hours**, at the upper end of what large-rotor turbines reach in North Frisia. Only the wind years were re-downloaded (the coordinates are part of the cache key).
- **Lesson:** a reanalysis value is a cell average. Always look at the grid point you got, its elevation, and whether it is land or sea. About 6% more wind speed gave 8% more energy.

4.3 The solar site earns a bit more than the national fleet

Capture rate	2023	2024	2025
Solar park (model)	0.781	0.622	0.555
National solar	0.759	0.592	0.516

Two candidate reasons, both plausible: (1) **one site is not the fleet**: the hourly correlation is 0.92, and when Lusatia is sunny while the rest of Germany is not, prices are higher; (2) **inverter clipping** removes energy exactly in the sunniest, cheapest hours. Separating them is a play idea in the notebook (change `dc_ac_ratio`). For pricing it means: the national capture rate is a good first guess, but a site's own profile can be worth a few EUR/MWh.

4.4 Small ones

- **Rate limits:** a whole year of hourly data counts as many API calls. With 4 s between requests, 64 site-years downloaded in 274 seconds without a single 429.
- **A test that tested formatting:** the first report test looked for `"| 2025 |"` in a Markdown table, which depends on column padding. It now checks that both years appear in the right section: test behaviour, not layout.
- **Constant series:** the synthetic national wind in the tests is constant, so its correlation is undefined; the function now returns NaN quietly instead of a warning.

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5. Hands-on: run it yourself

```
cd ".../Encavis quantitative/ppa-lab"
make test                  # 144 offline tests
make weather               # first run about 5 minutes (downloads), then seconds
open reports/step04_generation.md
make notebook              # Step 4 section: 4.1 to 4.6, then your solutions
```

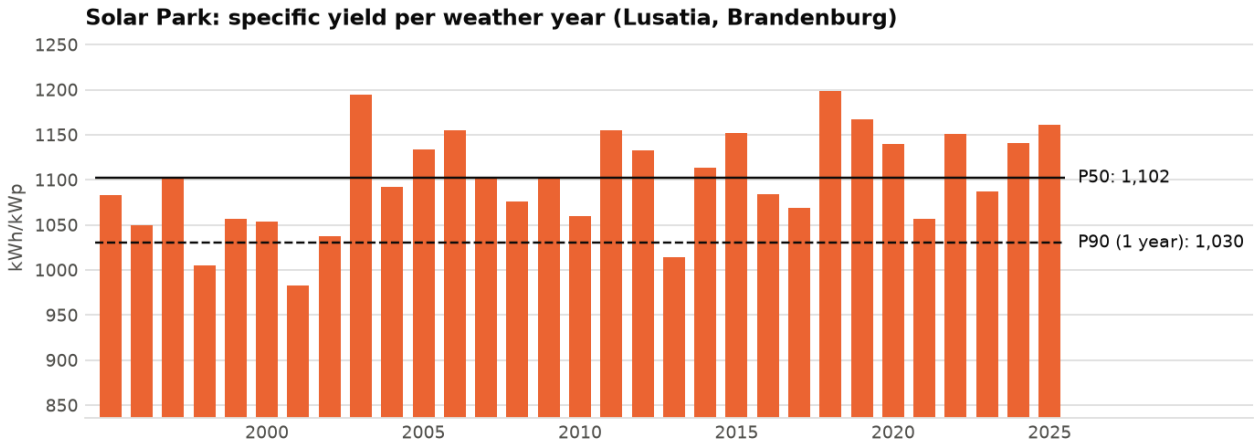
In the notebook, run 4.2 (the timestamp evidence yourself) and play with 4.3 (any week at both sites) and 4.6 (site vs national).

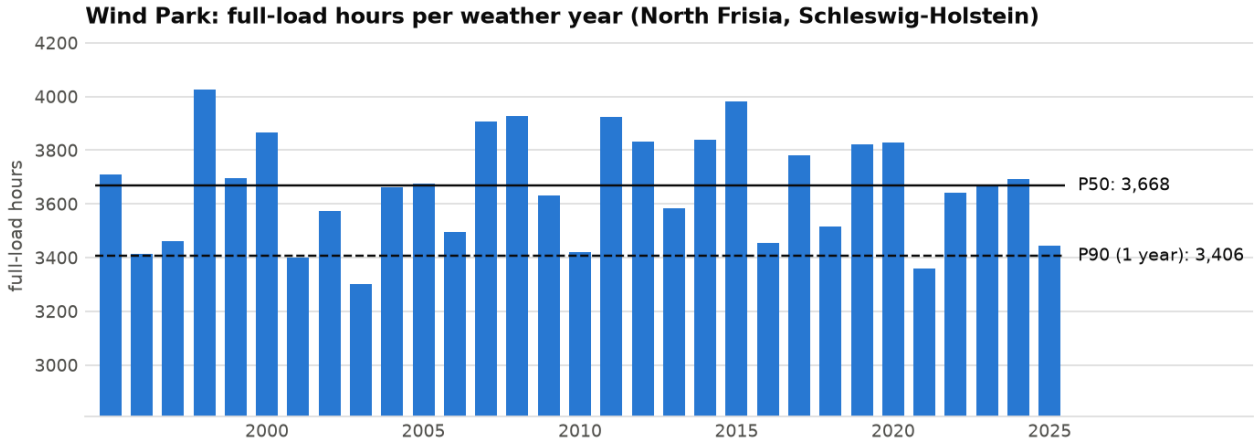
---

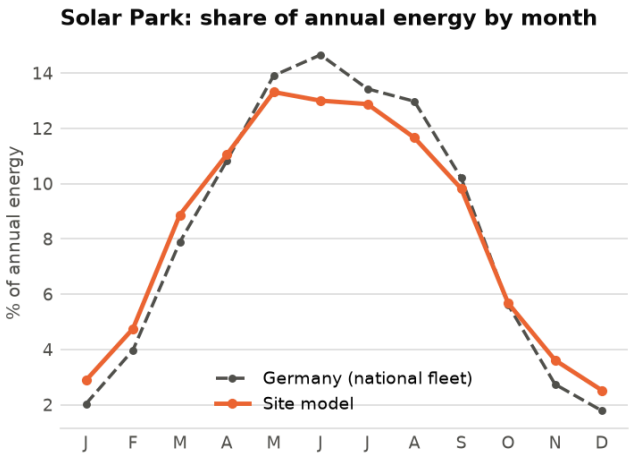
6. Results: what the data says

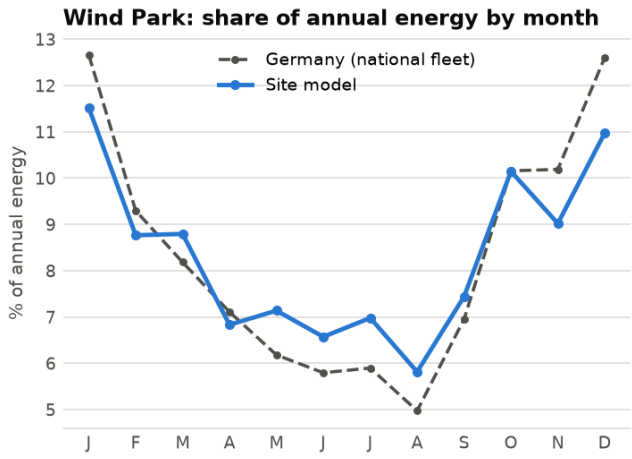
6.1 Volumes

	Solar park, Lusatia	**Wind park, North Frisia**
ERA5 grid point	51.75 N, 14.25 E, 68 m	54.50 N, 9.25 E, 30 m
P50	55.1 GWh = **1,102 kWh/kWp**	110.0 GWh = **3,668 full-load hours**
Variability (CoV)	5.0%	5.5%
P90 one year	51.5 GWh (–6.4%)	102.2 GWh (–7.0%)
P90 ten years	53.9 GWh (–2.0%)	107.4 GWh (–2.2%)
Worst / best year	2001 –10.7% / 2018 +8.9%	2003 –9.8% / 1998 +9.9%
Trend	+2.5% per decade	+0.1% per decade









6.2 Five stories

1. **One year in ten is 6-7% short.** That is the one-year P90 for both parks. For a 10-year average it is only about 2%: the reason long contracts can carry volume risk.
2. **Solar is getting sunnier.** +2.5% per decade over 1995-2025 matches the European "brightening" (cleaner air since the 1980s and 1990s). A P50 from the full 31-year mean may understate today's yields. Which period defines "standard weather" is a real decision, not a detail.
3. **The extremes are ±10%.** Encavis reports that annual wind yields can deviate by up to ±25%: real parks add outages, curtailment and shorter periods to the weather signal, which our potential-output model does not include. Treat our P90 as a weather-only floor.

4. **Wind is a winter product, solar a summer one.** Wind delivers 11.5% of its year in January and 5.8% in August; solar 13% in May and 2.5% in December. The two parks together are smoother than either alone (exercise 3 shows what that is worth).
5. **One site is not the fleet.** Hourly correlation with the national fleet is 0.92 for solar but only 0.72-0.75 for wind: wind volume risk is far more local.

6.3 What this means for pricing

- The **fair price** uses expected (P50) volumes; the **risk premium** and debt sizing use the spread (P90). Both now come from the site, not the country.
- Step 6 will pair these 31 weather years with simulated prices. Because windy and sunny hours are cheap hours, the volume and the price of each hour must stay together.

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7. Exercises

Warm-up (in `exercises/step04_generation.py`, checked by `.venv/bin/pytest exercises -q`)

1. **Specific yield and full-load hours** for both parks in 2024.
2. **P50 and P90** by hand from the 31 weather years.
3. **A solar + wind portfolio:** the capture rate of both parks sold together.
4. **Written:** one-year vs ten-year P90, and what makes the ten-year figure optimistic.

Core

5. **Clipping (test first).** Write a test that a DC/AC ratio of 1.0 produces more energy than 1.4 at the same DC capacity, then measure the energy and capture-rate difference for 2025. Does clipping explain the solar site's higher capture rate (section 4.3)?
6. **Hub height.** Re-run the wind model with 117 m and 157 m hubs for 2024. How much energy does each 20 m add, and why does the answer depend on the shear?
7. **Monthly P90.** Compute P90 for each calendar month of the solar park. Why do the twelve monthly P90s not add up to the annual P90?

Stretch

8. **Air density.** Add `surface_pressure` to the weather request and use windpowerlib's density correction. How large is the effect in winter vs summer?
9. **P99** with the normal approximation and empirically. Which do you trust with 31 years?

▼ **Hints and solution sketches**

1. `generation["mw"].tz_convert(TZ).loc["2024"].sum() / capacity`: about 1,141 kWh/kWp for solar and 3,692 hours for wind in 2024.
2. `annual.median()` and `annual.mean() - 1.2816 * annual.std(ddof=1)`.
3. Add the two `mw` Series, align with prices on common hours, then capture / mean price.
4. Averages of independent years have a standard deviation σ/√n. Lenders of a 10-year loan look at the 10-year number but add margins because years are not independent (trends, persistent

regimes) and because σ itself is estimated from only 31 years.
5. Call `solar_power_mw` twice with different `dc_ac_ratio`; compare sums and capture rates with Step 2's formula.
6. Higher hubs gain more where shear is strong (night, stable air).
7. Monthly deviations partly cancel within a year; adding monthly P90s adds twelve "bad months" that rarely happen together.
8. `ModelChain(..., density_model="barometric")` needs pressure and temperature at hub height.
9. `mean - 2.326 * std` vs the 1st percentile of 31 values (which is just the worst year).

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8. Self-check quiz

1. What is a reanalysis, and why use it instead of on-site measurements for long-term yields?
2. What are GHI, DNI and DHI?
3. Why is the sun position computed at mid-interval?
4. What did Open-Meteo's time label mean for radiation, and how did we check our conversion?
5. What is the DC/AC ratio, and what is clipping?
6. Why is the mean wind speed not enough to compute energy?
7. How is wind speed extrapolated from 100 m to hub height here?
8. Define P50, one-year P90 and ten-year P90. Why is the ten-year P90 closer to the mean?
9. Why did the first wind site give offshore-like full-load hours?
10. How do you check a model when no official figure exists?

▼ **Answers**

1. A weather model run over past decades that assimilates all observations, giving a consistent, gap-free hourly record everywhere; sites rarely have decades of measurements.
2. Global horizontal (everything on a flat surface), direct normal (the sun's beam on a surface facing it), diffuse horizontal (scattered sky light).
3. An hourly mean describes the whole hour; the sun at its middle represents it best.
4. Mean of the preceding hour (labelled at the end). After shifting, irradiance must match the clear-sky sun best at +30 minutes; a test and a build check enforce it.
5. Panel (DC) capacity divided by inverter (AC) capacity; above the inverter limit output is cut off, which loses some energy in the sunniest hours.
6. Power rises roughly with the cube of speed and is capped and cut out by the power curve, so the distribution of speeds matters, not just its mean.
7. Power law with the shear exponent measured each hour from the 10 m and 100 m speeds: v_hub = v100 · (h/100)^alpha, alpha clipped to [0, 0.6].
8. Median year; mean − 1.2816σ; mean − 1.2816σ/√10. Averaging ten independent years shrinks the standard deviation by √10.
9. Its ERA5 grid cell had 0 m elevation, partly sea, so the cell's wind was offshore-like.
10. Benchmarks, hourly correlation and seasonal shape against the national fleet, capture rate at real prices, and physical checks such as timestamps against the sun.

9. Interview corner

Your 60-second story (STAR):

> *Situation:* to price a PPA and size its debt, a bank needs the park's own P50 and P90, not a national profile. *Task:* model hourly output of a 50 MWp solar and a 30 MW wind park at real German sites over many weather years. *Action:* I pulled 31 years of ERA5 reanalysis for each site, converted its end-of-hour radiation labels and verified them against the sun's position, modelled the parks with pvlib and windpowerlib, and checked them against the national fleet. *Result:* 1,102 kWh/kWp and 3,668 full-load hours at P50, one-year P90s about 6-7% below the mean, and two traps caught on the way: shifted timestamps, and a wind site whose grid cell was partly sea.

Likely questions and strong answers:

- "*How do you use weather data in risk?*" Long-term reanalysis (ERA5) at the site, a physical model to power, annual energies per weather year for P50/P90, and the same weather years paired with prices for joint risk, because volume and price are correlated.
- "*Your model and the real park disagree. What do you check first?*" Timestamps and time zones, the grid point and its surroundings, the power curve and hub height, losses and curtailment, then the long-term period used.
- "*What drives the difference between one-year and ten-year P90?*" Averaging: $\sigma/\sqrt{n}$ under independence. In practice lenders add margins for trends, persistence and estimation error.
- "*Is wind or solar riskier on volume?*" Here similar year to year (5.0% vs 5.5%), but wind is far more local (hourly correlation with the fleet 0.72 vs 0.92) and more sensitive to modelling choices such as hub height and site exposure.

10. How this maps to real work at Encavis

In this project	In a company
ERA5 via Open-Meteo	ERA5, MERRA-2 and commercial reanalyses; on-site met masts and lidar
pvlib / windpowerlib models	Energy yield assessments by consultants; SCADA data of operating parks
P50/P90 over 31 weather years	Budget (P50), debt sizing (P90/P99), "standard weather" over 30 years
Sun check on timestamps	Data-quality rules on SCADA and meter data
Grid cell in the sea	Site exposure and representativeness checks in yield reports
Site vs national capture rate	Asset-level capture-rate monitoring and PPA settlement checks

Encavis reports that annual wind yields can deviate strongly from the mean and uses long-term data, external yield reports and sensitivity analyses: today's work is the small, transparent version of that toolbox.

11. Your to-dos for Step 4

- □ Read sections 1 and 4; redo the shear calculation in 1.4 for v10 = 5, v100 = 7.5 m/s and a 137 m hub (you should get about 7.93 m/s).
- □ Run section 5; in the notebook, reproduce the timestamp table of section 4.1 (cell 4.2).
- □ Exercises 1-4 in `exercises/step04_generation.py`; check them in "My solutions: Step 4".
- □ **Commit Step 4** (commands below).
- □ **German vocabulary of this step:** die Globalstrahlung, die Direktstrahlung, die Diffusstrahlung, die Modultemperatur, der Wechselrichter, die Nabenhöhe, die Leistungskennlinie, die Volllaststunden, der spezifische Ertrag, das Windjahr, der Nachlaufverlust (wake loss), die Verfügbarkeit, die Reanalyse.
- □ Stand-up update: "*Yesterday: site models for the solar and wind parks from 31 years of ERA5; P50 1,102 kWh/kWp and 3,668 full-load hours, P90 one year about −6 to −7%. Today: stochastic price model (PPA-5). Blockers: none.*"

Commit (check `git status` first: no `.env`, no `M .env.example`):

```
cd ".../Encavis quantitative/ppa-lab"
git status --short
git add src tests exercises config reports docs notebooks METHODOLOGY.md README.md
CHANGELOG.md Makefile pyproject.toml
git status --short
git commit -m "feat(weather): ERA5 site models for solar and wind, P50/P90 volumes (PPA-4)"
git push
gh run watch
```

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12. Glossary

Term	Meaning
Reanalysis	Weather model run over the past, assimilating observations (ERA5)
GHI / DNI / DHI	Global horizontal / direct normal / diffuse horizontal irradiance
POA irradiance	Irradiance on the plane of the tilted panels
DC/AC ratio, clipping	Panel vs inverter capacity; output cut at the inverter limit
Specific yield	Annual energy per kWp of panels (kWh/kWp)
Wind shear, alpha	Increase of wind speed with height; power-law exponent
Power curve	Turbine output as a function of hub-height wind speed
Full-load hours	Annual energy divided by capacity
Wake loss	Energy lost because turbines shadow each other
P50 / P90	Energy exceeded in 50% / 90% of years
Weather year	One historical year of weather run through the park model
Grid point / cell	Location and area a reanalysis value represents

13. Sources

- Hersbach, H. et al. (2020): The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146, 1999-2049.
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